William Lim

William.Lim@csu.fullerton.edu | walofcode.com | github/FrewtyPebbles | in/william-lim-87733a270

EDUCATION

California State University, Fullerton **May 2026**

B.S. in Computer Science, B.A. in Studio Art

Relevant Coursework: Compilers & Languages, Operating Systems, Assembly, Computer System Architecture, Python, Scientific Writing, Algorithm Engineering, Back-End Engineering, Cybersecurity, File Structures & Databases, Multivariable Calculus, Statistics, Physics, Artificial Intelligence, Machine Learning, Edge AI, Software Architecture, Modern Software Deployment and Operations, Linear Algebra, Discrete Math, Design, Illustration

TECHNICAL SKILLS

Languages: Python, C, C++, Java, C#, Intel x86 and RISC-V assembly, TypeScript, HTML/CSS, SQL, R, Prolog, PHP, Rust

Frameworks, Skills & Tools: AWS (serverless/autoscaling, caching, security, events, etc.), Terraform (and Packer), React, Next.js, Node.js, Prisma, SQLAlchemy, vector databases, FastAPI, Flask, Docker, GitHub Actions, PyTorch, (Geo)Pandas, Matplotlib, NumPy, Scikit-Learn, OpenCV, Linux, OpenGL, WebGL2, Media Pipe, Ollama, Gemini API, SEO, MCP, Coverage and Unit Testing, SSH, Firewalls, Least Privilege, AES, RSA, Wire Shark, Scrum, Kanban, MongoDB, DynamoDB, PHP, Apache

EXPERIENCE

Web Designer, *Ride With Care Ambulette Transportation* **Aug. 2024 – Jan. 2025**

Designed public facing website UI and acted as SEO and IT consult.

Web Designer, *ExecPathfinders LLC* **July 2023 – Oct. 2024**

Designed public facing website UI and acted as SEO and IT consult. Integrated Google Analytics. Acted as on call technical consult during Amazon best seller book launch.

Software Engineer Intern, *Garrett Integrated Automotive Corp. (GIAC)* **Jan. 2024 - Jan. 2024**

Developed proprietary C++ dynamic-link library for querying legacy ISAM databases.

June 2022 - Aug. 2022

Developed REST API for product data queries. Developed data portal for customer support and a reactive embed for business partners' websites. Optimized/hardened SQL queries. Fixed AES CBC encryption bugs in C++ codebase.

PROJECTS | *Code can be found on GitHub.*

RISC-V CPU Emulator and Assembler **Nov. 2025**

Written in Python, it emulates a subset of RV32I and RV32F using bit-level logic gates. Includes RISC-V assembler.

A-Star GPS Road Navigation System **Dec. 2025**

Using A* and UCS on OpenStreetMap data, this GPS software navigates optimally to various geological coordinates. The system is written in python and accounts for traffic laws, road junctions, signage, and lane counts.

Vanta 3D Engine **Jan. 2026 – Apr. 2026**

Portable 3D engine using WebGL2, C++ (web assembly), and TypeScript. Includes Node/Scene system and PBR pipeline.

Python to Intel X86 Assembly Compiler and Bindings **Dec. 2024 - Sep. 2025**

Compiles Python code into statically typed x86 Assembly by compiling Python ASTs. Includes x86 Assembly bindings.

UDP Multi-Hop Mesh Network Protocol **Jun. 2026 - Present**

1 day python project still being extended/developed, uses BFS to get optimal paths for packet streaming in P2P mesh.

LEADERSHIP EXPERIENCE

Board Member, Association for Computing Machinery (ACM) Algorithms **Aug. 2025 – Mar. 2026**

Board Member, Association for Computing Machinery Game Development **Aug. 2025 – Jan. 2026**

California State University, Fullerton

- Prepared and presented lectures to Computer Science students on basic to advanced CS concepts.
- Organized, led, and performed outreach attracting new members to ACM, the largest CSUF student SWE organization.

Participant/Group Leader, UCLA Glitch Club Build with Gemini Google Hackathon **Mar. 2026**